Socioeconomic Predictors of Urban Crime in Europe

A Comparative Panel Data Analysis (2015–2022)

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Abstract

This study investigates the relationship between socioeconomic conditions and intentional homicide rates across six Western European countries — Portugal, Spain, France, Germany, Italy, and the Netherlands — using panel data from 2015 to 2022. Data were sourced from Eurostat and the World Bank, covering crime rates, unemployment, GDP per capita, income inequality (Gini index), and urban population share. A pooled ordinary least squares (OLS) regression yields an R-squared of 0.071, indicating that unemployment and GDP per capita alone explain only a small fraction of cross-national variation in crime rates at the aggregate level. However, within-country correlation analysis reveals substantial heterogeneity: Italy and Portugal exhibit strong positive correlations between unemployment and crime (+0.82 and +0.70, respectively), while the Netherlands shows a negative correlation (−0.63). These findings reveal Simpson's Paradox and underscore the importance of country-level institutional and cultural mediators. The study contributes to the growing field of computational social science by demonstrating how open data and statistical programming can illuminate complex social phenomena.

Keywords: crime, unemployment, inequality, panel data, computational social science, Europe, Simpson's Paradox

1. Introduction

The relationship between economic conditions and crime has been a central concern of sociological and criminological research for decades. Classical strain theory (Merton, 1938) posits that economic hardship creates pressure toward criminal behaviour, while social disorganization theory (Shaw & McKay, 1942) links crime to neighbourhood-level socioeconomic deprivation. More recently, the field of computational social science has enabled researchers to test these theories using large-scale, cross-national panel data and statistical programming techniques.

This study examines the extent to which socioeconomic factors — specifically unemployment, GDP per capita, income inequality (Gini index), and urban population share — explain variations in intentional homicide rates across six Western European countries from 2015 to 2022. The countries selected — Portugal, Spain, France, Germany, Italy, and the Netherlands — represent diverse socioeconomic profiles within the European context, ranging from Southern European economies with higher unemployment and inequality to Northern European economies with stronger social safety nets.

The central research question guiding this analysis is: To what extent do socioeconomic factors explain variations in crime rates across European countries over time?

Four hypotheses are tested:

H1: Higher unemployment is associated with higher crime rates.
H2: Higher income inequality (Gini index) predicts higher crime.
H3: Higher GDP per capita is associated with lower crime.
H4: Urban density increases crime concentration.

The study employs a mixed-methods quantitative approach, combining pooled OLS regression with within-country correlation analysis. The divergence between aggregate and country-level results reveals a Simpson's Paradox, highlighting the methodological importance of disaggregated analysis in comparative social research.

2. Data and Methods

2.1 Data Sources

All data used in this study are publicly available from official statistical agencies. Crime data were obtained from Eurostat's crim_hom_soff dataset, which records intentional homicide rates per 100,000 inhabitants. Unemployment rates were sourced from Eurostat's tipsun20 dataset, defined as the percentage of the labour force aged 15 to 74 that is unemployed. GDP per capita in current US dollars, the Gini index of income inequality, and urban population as a percentage of total population were obtained from the World Bank's open data portal.

2.2 Country and Time Selection

Six countries were selected to represent diverse socioeconomic conditions within Western Europe: Portugal, Spain, France, Germany, Italy, and the Netherlands. The time period covers 2015 to 2022, providing eight years of balanced panel data. This period captures post-financial crisis recovery, the COVID-19 pandemic, and subsequent economic fluctuations.

2.3 Variables

The dependent variable is intentional homicide rate per 100,000 inhabitants. The independent variables are: unemployment rate (%), GDP per capita (current US$), Gini index (0–100 scale), and urban population (% of total). All variables are measured at the national level on an annual basis.

2.4 Analytical Strategy

The analysis proceeds in three stages. First, exploratory data analysis is conducted through descriptive statistics, correlation matrices, and data visualisation including bar charts, time series plots, and scatter plots. Second, a pooled ordinary least squares (OLS) multiple regression is estimated with crime rate as the dependent variable and unemployment and GDP per capita as independent variables. Third, within-country Pearson correlation coefficients are calculated between crime and unemployment for each of the six countries individually. All analysis was performed using Python (pandas, numpy, matplotlib, seaborn, statsmodels).

2.5 Limitations

Several limitations should be acknowledged. Crime definitions and recording practices vary across countries, which may affect comparability. The analysis uses national-level data, which masks within-country regional variation. Missing data for certain years, particularly Portugal's homicide rates before 2019 and the Gini index for the Netherlands in 2022, reduce the sample for specific analyses. The pooled OLS model does not control for unobserved country-specific heterogeneity. Finally, correlation does not imply causation; the relationships observed may be influenced by confounding variables not included in the model.

3. Results

3.1 Descriptive Statistics

Table 1 presents summary statistics for all variables across the 48 country-year observations. The average intentional homicide rate across the sample is 0.85 per 100,000 inhabitants, ranging from 0.48 (Italy, 2020) to 1.55 (France, 2015). Unemployment averages 8.81%, with substantial variation from 2.90% (Germany, 2019) to 22.10% (Spain, 2015). GDP per capita averages $37,395, ranging from $19,216 (Portugal, 2015) to $60,142 (Netherlands, 2021). The average Gini index is 32.76, and the average urban population share is 77.14%.

3.2 Crime Rates by Country

France consistently records the highest average homicide rate (1.16 per 100,000), followed by Germany (0.91), Portugal (0.76), the Netherlands (0.68), Spain (0.65), and Italy (0.59). Time series plots reveal that crime rates follow distinct national trajectories rather than a common European trend, with France showing the greatest volatility.

3.3 Correlation Analysis

The correlation matrix shows that crime rate has weak correlations with all socioeconomic variables at the aggregate level: unemployment (−0.27), GDP per capita (+0.20), Gini index (−0.21), and urban population (+0.07). In contrast, the socioeconomic variables are strongly intercorrelated. GDP per capita is negatively correlated with unemployment (−0.69) and the Gini index (−0.80), and positively correlated with urban population (+0.78).

3.4 Pooled OLS Regression

The pooled OLS regression of crime rate on unemployment and GDP per capita yields an R-squared of 0.071 (adjusted R-squared: 0.026), indicating that the model explains only 7.1% of the variation in crime rates. Neither unemployment (coefficient = −0.013, p = 0.220) nor GDP per capita (coefficient ≈ 0.000, p = 0.968) is statistically significant at conventional levels. The F-statistic is 1.573 (p = 0.220), confirming that the model as a whole is not statistically significant.

3.5 Within-Country Correlations

The within-country analysis reveals a pattern entirely obscured by the pooled model. Italy (+0.82) and Portugal (+0.70) exhibit strong positive correlations between unemployment and crime, consistent with H1. France (+0.49) and Germany (+0.36) show moderate positive correlations. However, Spain (−0.29) and the Netherlands (−0.63) display negative correlations, indicating that crime rates in these countries moved in the opposite direction from unemployment during the study period. This divergence between aggregate and disaggregated results constitutes a Simpson's Paradox.

4. Discussion

The findings of this study challenge simplistic narratives about the relationship between economic conditions and crime. At the aggregate level, the pooled regression model explains very little of the cross-national variation in homicide rates, and neither unemployment nor GDP per capita emerges as a statistically significant predictor. This null finding at the aggregate level might lead a naive analyst to conclude that socioeconomic factors are irrelevant to crime.

However, the within-country correlation analysis tells a different and more nuanced story. In Southern European countries — Italy and Portugal — the relationship between unemployment and crime is strongly positive, as classical criminological theories would predict. When unemployment falls, crime tends to fall alongside it. This pattern is consistent with strain theory: improved economic conditions reduce the pressure toward criminal behaviour. It also aligns with the historical experience of these countries, where periods of economic crisis have been associated with social instability.

In contrast, the Netherlands shows a strongly negative correlation: crime rates have remained relatively stable or even increased slightly while unemployment has declined. This counterintuitive finding may reflect the mediating role of robust social institutions, comprehensive welfare systems, and effective labour market policies that decouple economic cycles from social dysfunction. It is also possible that crime in these countries is driven more by non-economic factors such as demographic change, drug markets, or shifts in policing practices.

The divergence between pooled and within-country results constitutes a classic Simpson's Paradox. When data from heterogeneous countries are aggregated, the opposing within-country dynamics cancel each other out, producing a misleading null result. This finding has important methodological implications for comparative social research: cross-national analyses that do not account for country-level heterogeneity risk drawing erroneous conclusions. It also has policy implications: one-size-fits-all approaches to crime prevention based on economic policy are unlikely to be effective. Interventions must be tailored to national institutional contexts.

The weak overall explanatory power of the model (R-squared = 0.071) also suggests that factors beyond the scope of this study — such as institutional quality, social spending, policing strategies, drug markets, and cultural attitudes toward violence — may be more important determinants of crime than the macroeconomic variables examined here. This is consistent with a growing body of literature emphasising the multidimensional nature of crime causation.

5. Conclusion

This study set out to examine the relationship between socioeconomic conditions and crime across six European countries using panel data from 2015 to 2022. The results reveal a complex picture: while aggregate-level analysis finds no statistically significant relationship between unemployment or GDP and crime rates, within-country analysis uncovers substantial heterogeneity. The discovery of a Simpson's Paradox — where opposing national patterns cancel each other out at the aggregate level — constitutes the study's key empirical contribution.

These findings carry implications for both research methodology and policy. Methodologically, they demonstrate the dangers of pooled cross-national analysis without adequate attention to country-level heterogeneity. Future research on crime and socioeconomic conditions should routinely disaggregate by country and incorporate country-specific institutional variables. From a policy perspective, the results caution against uniform, economically deterministic approaches to crime prevention.

The study also serves a pedagogical purpose. As an undergraduate project in Sociology transitioning toward computational social science, it demonstrates how open data from Eurostat and the World Bank, combined with Python-based statistical analysis, can produce meaningful empirical contributions. The complete code, data, and interactive dashboard are publicly available in the accompanying GitHub repository, reflecting a commitment to transparency and reproducibility.

6. Future Research Directions

Several extensions of this work are planned or underway. First, a fixed-effects panel regression model will be estimated to control for unobserved time-invariant country characteristics, providing more robust causal estimates than the pooled OLS approach presented here. Second, the analysis will be expanded to include Eastern European countries, increasing sample size and variation. Third, spatial regression models incorporating GIS data will test whether geographic factors mediate the relationship between socioeconomic conditions and crime. Fourth, social media sentiment analysis will be incorporated as a predictor variable, testing whether public mood anticipates changes in crime rates. Fifth, causal inference methods such as difference-in-differences will be applied to specific policy interventions. These planned extensions will progressively strengthen the analytical depth of the project.

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